计算理论

THEORY OF COMPUTATION

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布尔电路的计算能力

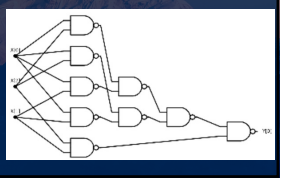
- 布尔电路的计算能力

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NAND电路与{AND,OR,NOT}电路等价

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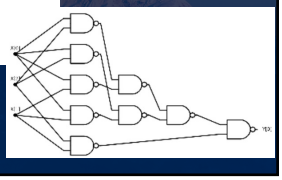
用NAND电路计算MAJ:$\{0,1\}^3 \to \{0,1\}$



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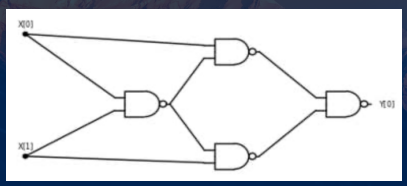
用NAND电路计算MAJ:$\{0,1\}^3 \to \{0,1\}$

$$NAND\Big(NAND\Big(NAND(NAND(a,b),NAND(a,c)),$$
$$NAND(NAND(a,b),NAND(a,c))\Big),$$
$$NAND(b,c)\Big)$$



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用NAND电路计算XOR



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▶一个大小为s的电路与一个规模至多为3s的NAND电路等价

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▶一个大小为s的电路与一个规模至多为3s的NAND电路等价

- $NOT(a) = NAND(a, a)$
- $AND(a, b) = NAND(NAND(a, b), NAND(a, b))$
- $OR(a, b) = NAND(NAND(a, a), NAND(b, b))$

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两个**计算模型等价**，当且仅当它们计算的“函数集合”是相同的

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布尔电路是一个好的计算模型吗?

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好模型?



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好模型?





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好的模型

用模型M描述一个对象集合X

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好的模型

用模型M描述一个对象集合X

- 逼真度(fidelity)
  - 假设我们想得到X上的“问题集合”Q的答案，M应该有如下性质

$$\forall q \in Q, \forall x \in X, q(\mathrm{M}(x)) \approx q(x)$$

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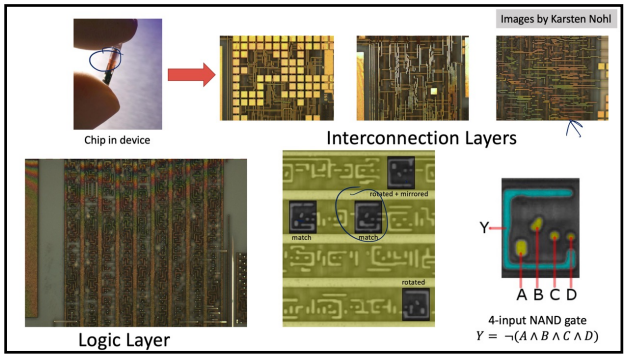
- 简单
  - M(x)应该比x更容易描述、理解
- 完备性(completeness)

$$\forall x \in X, \mathrm{M}(x)\text{都存在}$$

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布尔电路是一个好的计算模型吗?

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Universality

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将gate扩展为任意可能的布尔函数

A gate set, G, is **universal**, if for every Boolean (AND,OR,NOT) circuit, there is a Boolean (G) circuit that computes the same function.

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如何做universality的相关证明

- （1）证明G是universal的
  - 用G模拟AND、OR和NOT
- （2）证明G不是universal的
  - 找到一个函数f，f可以被AND、OR、NOT计算，但无法被G计算

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证明：{OR}不是universal的

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证明：{OR,NOT}是universal的

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证明：{AND,OR}不是universal的

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▶证明{NAND}是universal的

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为什么universality用{AND, OR, NOT}定义?

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可以证明，任意形如$\{0,1\}^n \to \{0,1\}$的函数，都可以用{AND, NOT, OR}计算

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▶先来看一个特别的gate

▶$G = \{IF, ZERO, ONE\}$

▶$ZERO: \{0,1\} \to \{0\}$，$ONE: \{0,1\} \to \{1\}$

▶$IF(a,b,c) = \begin{cases} b & \text{如果} a = 1 \\ c & \text{如果} a = 0 \end{cases}$

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- 先来看一个特别的gate
  - $G = \{IF, ZERO, ONE\}$
  - $ZERO: \{0,1\} \to \{0\}$，$ONE: \{0,1\} \to \{1\}$
  - $IF(a,b,c) = \begin{cases} b & \text{如果} a = 1 \\ c & \text{如果} a = 0 \end{cases}$
- G是universal的

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证明任意形如$\{0,1\}^n \to \{0,1\}$的函数，都可以用{AND, NOT, OR}计算

$$LOOKUP_k\left(t_1, \dots, t_{2^k}, x_1, \dots, x_n\right) \to \{0,1\}$$

0	0	⋯	0	0	0	t_1
0	0	⋯	0	1	1	
0	0	⋯	0	0	0	
⋮	⋮	⋯	⋮	⋮	⋮	
1	1	⋯	1	1	1	t_{2^k}

$\{0,1\}^n \to \{0,1\}^m$ for $n, m \in \mathbb{N}$

$$LOOKUP_k\left(t_1^1, \dots, t_{2^k}^1, x_1, \dots, x_n\right) \to \{0,1\}$$
$$LOOKUP_k\left(t_1^2, \dots, t_{2^k}^2, x_1, \dots, x_n\right) \to \{0,1\}$$
$$\dots$$
$$LOOKUP_k\left(t_1^m, \dots, t_{2^k}^m, x_1, \dots, x_n\right) \to \{0,1\}$$

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- 定理：任意形如$f: \{0,1\}^n \to \{0,1\}^m$的函数可以被包含至多$c \cdot m2^n$逻辑门的布尔电路计算。

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- 每一个有穷函数(输入输出数量有穷)可以被足够大的布尔电路计算

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- 我们并不需要过于复杂的逻辑门
- 很多逻辑门看起来能力更强，但实际不是
- 剩下的问题是，如何改进效率

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- 如何衡量效率?

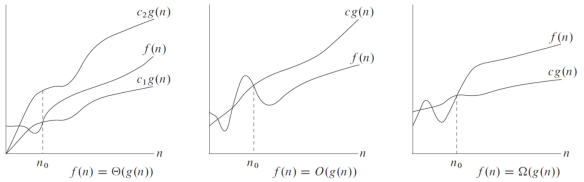
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Informal meaning of $O, \Omega, \Theta, o, \omega$

- For each $\alpha \in \{O, \Omega, \Theta, o, \omega\}$, and any (usually non-decreasing) function $g(n) : \mathbb{N} \to \mathbb{R}^+$ the notation $\alpha(g(n))$ denotes a **set of functions that are related to** $g(n)$
- We write $f(n) = \alpha(g(n))$ but what we mean is $f(n) \in \alpha(g(n))$.

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Pictorially:



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Formal definition of O, Ω, Θ

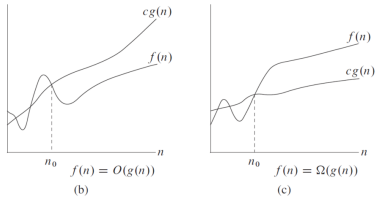
$O(g(n)) = \{f(n) :$ there exist positive constants c and n_0 such that $0 \le f(n) \le cg(n)$ for all $n \ge n_0\}$.

$\Omega(g(n)) = \{f(n) :$ there exist positive constants c and n_0 such that $0 \le cg(n) \le f(n)$ for all $n \ge n_0\}$.

For any two functions $f(n)$ and $g(n)$, we have $f(n) = \Theta(g(n))$ if and only if $f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$. ■

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Informal picture of ω, o



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Formal definitions of ω, o

$o(g(n)) = \{f(n) :$ for any positive constant $c > 0$, there exists a constant $n_0 > 0$ such that $0 \le f(n) < cg(n)$ for all $n \ge n_0\}$.

$\lim_{n\to\infty} \frac{f(n)}{g(n)} = 0$

$\omega(g(n)) = \{f(n) :$ for any positive constant $c > 0$, there exists a constant $n_0 > 0$ such that $0 \le cg(n) < f(n)$ for all $n \ge n_0\}$.

$\lim_{n\to\infty} \frac{f(n)}{g(n)} = \infty$

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Examples

- $n = O\left(\frac{n}{2}\right), n = O(n^2)$
- $\Omega(\cdot)$ is the inverse of $O(.)$
- Θ : when $O(.)$ and $\Omega(\cdot)$ both happen → example: $2n^2 = \Theta(n^2 + 100\, n)$
- $n = o(n^2)$
- ω is the inverse of $o(\cdot)$ → example $2^n = \omega(n^{10})$
- $\log(n!) = \Theta(n \log n)$

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▸效率, Circuit Complexity

We can build the circuit! But how many gates are needed?

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